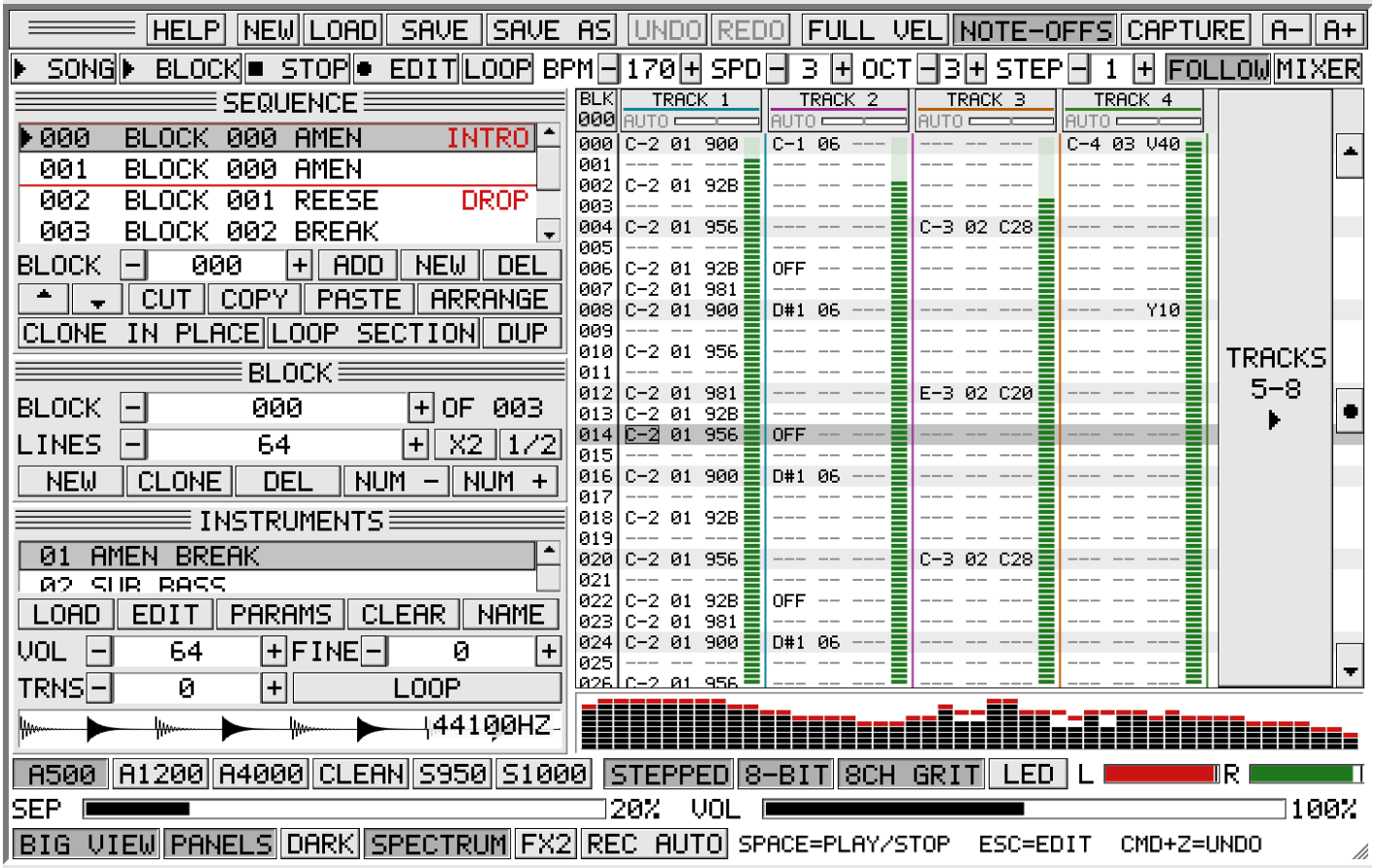
PAULA'S REVENGE V0.23

USER MANUAL

AN 8-TRACK AMIGA-STYLE SAMPLE TRACKER

FOR JUNGLE, DRUM & BASS AND BREAKS



THE MAIN SCREEN, LIGHT SKIN

Paula's Revenge is an original program inspired by the classic Amiga trackers of the early 90s. Paula and Akai sound, a mixer, plug-ins, MIDI out, automation, bouncing and stems.

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GETTING STARTED

Paula's Revenge is an 8-track sample tracker for jungle, drum & bass and anything else that wants the crunchy sound of the Amiga's Paula chip and the old Akai samplers. You write music as lines of notes and commands that play from top to bottom, one block at a time, in the order the song lists them.

INSTALL

1. Open **Releases** on the right of the GitHub repo page. Each release has both versions: `PaulasRevenge-macOS.zip` for a Mac and `PaulasRevenge-Windows.zip` for a PC (Windows 10 or 11, 64-bit).
2. **Mac:** follow `INSTALL.txt` inside: copy the app into Applications and clear the macOS quarantine.
   PC: unzip it anywhere and run `Paula's Revenge.exe`. The first time, Windows says *Windows protected your PC* because the app isn't code-signed yet: click **More info**, then **Run anyway**.
3. Open it. Audio and MIDI devices are under **Audio/MIDI Settings…** (CMD ,) in the **File** menu (on a Mac also in the app menu). The File menu has New, Load…, Save, Save As…, **Save MOD…**, **Render WAV…** and **Export Samples…**; the View menu has Capture, Sample Audio, **Record Note-Offs**, **Full Velocity** and full screen.
4. The sound follows the computer's output: plug in headphones or connect AirPods and it moves there, with the status line saying where it went. **Follow System Output (Headphones)** in the menu turns this off; picking a device by hand in Audio/MIDI Settings turns it off too, so your choice stays.

> **PC** Every CMD shortcut in this manual is CTRL on a PC, and the app shows CTRL where the Mac shows CMD. Full screen is F11. Plug-ins are VST3 (Audio Units are Mac-only).

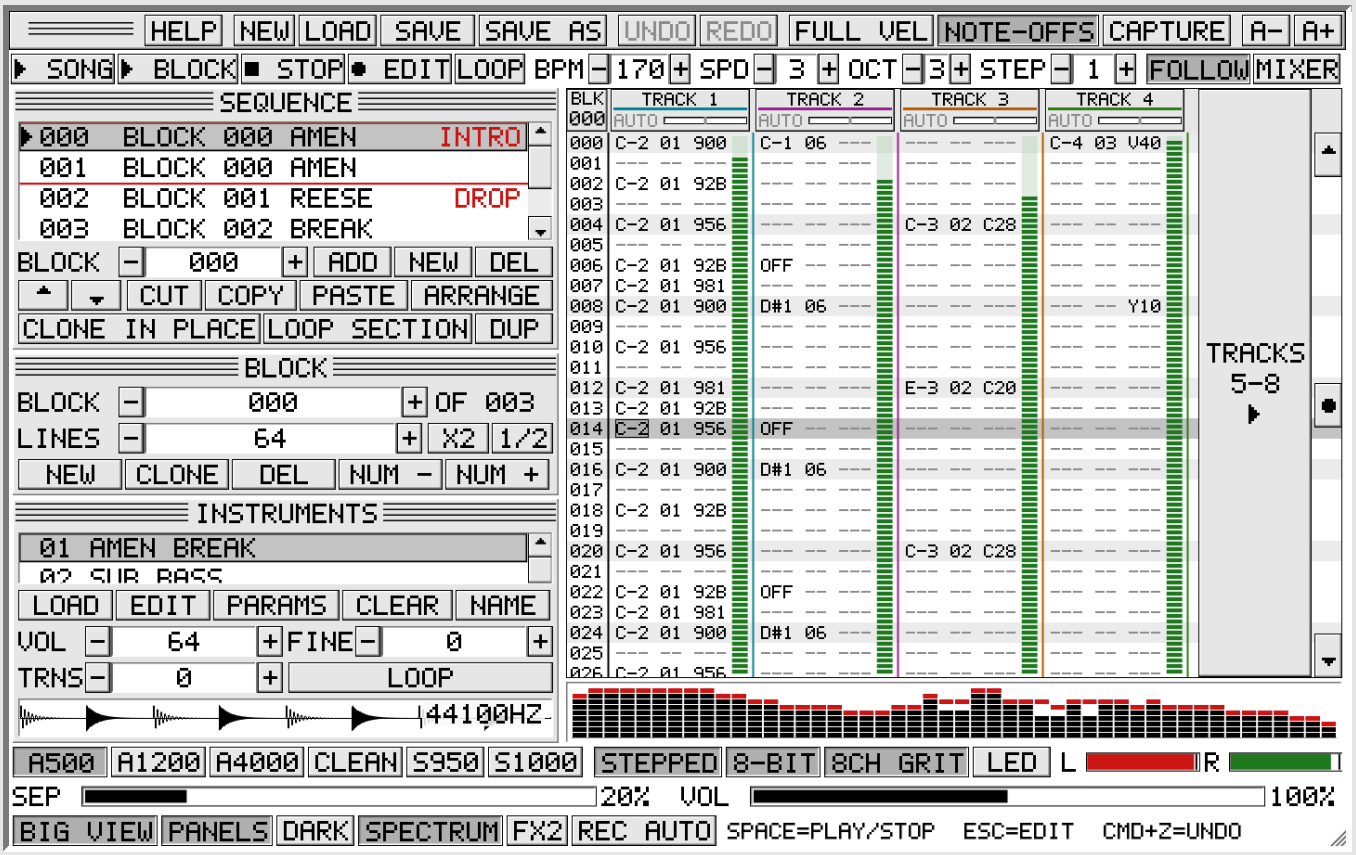
YOUR FIRST SOUND IN A MINUTE

1. Press SPACE. A new song already has a few chip sounds in it, so the keys make sound straight away.
2. Press ESC to switch • EDIT on. The cursor line turns grey.

3. Type notes with the bottom two rows of the keyboard (`Z` is C, `S` is C#, `X` is D…). Each note goes in with the current instrument and the cursor moves down by `STEP`.
4. Click `LOAD` in the instrument panel to pick a break from your drive. Every sample you highlight plays straight away.
5. Press `SPACE` to hear the block. `SHIFT` `SPACE` plays the whole song.

> **TIP** For a break: put `C-3` on line 0 and use `9xx` offsets to jump around inside it, or slice it in the sample editor and play the slices from the keys.

THE SCREEN



THE MAIN SCREEN, LIGHT SKIN, PLAYING

AREA	WHAT IT DOES
Title bar	NEW, LOAD, SAVE, SAVE AS, UNDO, REDO, FULL VEL, NOTE-OFFS, CAPTURE (see Recording), and text size A- A+. Save MOD, Render WAV and Export Samples are in the **File** menu.
Transport	Play the song or the block, stop, EDIT mode, LOOP (the marked lines), BPM, SPD (ticks per line), octave and step, FOLLOW (CMD F) and MIXER (CMD 2).
Sequence	The song order. Click a position to go there, drag to highlight positions, and set which block plays where.
Block	Pick, add, clone or delete blocks; set the number of lines (1–256); X2 doubles a block, 1/2 halves it.
Instruments	63 slots. LOAD, EDIT, PARAMS, the instrument's VOL, finetune, transpose, loop, and a little waveform.
Pattern	8 tracks. Click a track name to mute it, right-click to solo; drag in the bar under the name to pan the track.

AREA	WHAT IT DOES
Output strip	The machine you hear (A500, A1200, A4000, CLEAN, S950, S1000), its options, an L/R meter, stereo separation and volume.
Bottom row	BIG VIEW, PANELS, DARK, SPECTRUM, FX2, REC AUTO, and the status line.

The app opens with the dark skin. DARK switches to black on white, as in this manual. Every song has 8 tracks (a 4-channel MOD fills tracks 1–4). BIG VIEW shows 4 at a time with large text, with a tall TRACKS 5-8 ▶ tab beside them; click it for tracks 5–8, and ◀ TRACKS 1-4 brings you back (TAB past track 4 flips too). The song's name (in the window's title bar, and at the top when there's room) is its file's name: save it as `amen roller.prv` and it's AMEN ROLLER.

The status line at the bottom right says what just happened, and what the command under the cursor does.

RIGHT-CLICK MENUS

Right-click almost anything for the things you can do to it. Everything they do can be undone.

RIGHT-CLICK	MENU
A note in the pattern	Copy / cut / paste the track, clear the line, insert or delete a line, effect commands, VU meter colour, spectrum colour
An effect cell	The command list; on a `#xy`, the chord list
A highlight in the pattern	Bounce it to a new sample
An instrument	Copy, paste, duplicate to the next free slot, clear, rename, load a sample, sample (record audio), play `C-3`
A song position (sequence or ARRANGE)	Copy, cut, paste, repeat, delete, insert the block again, section label
The block panel	Copy / cut / paste / clear the block, duplicate, new, delete, double, halve
The sample's waveform	The sample editor's edits on the highlight

RIGHT-CLICK	MENU
A mixer strip's name	Copy / paste / reset the strip, open the channel
A mixer effect slot	Copy / paste the effect, clear the slot, bypass, open
A CHORDS slot	Copy, paste, clear (all the picked slots), colour, play, record chords, note
Any level meter: the bar at the right of each track, the L / R meter, the mixer's meters	Their colour, one for all of them: green, amber, cyan, red, blue, pink, cream / black, or each track's colour (kept for next time)
The spectrum strip	Its colour: the same choices, or rainbow (the track colours, as it starts)
EDIT	Recording: note-offs in step entry, detect chords, capture, NOTE-OFFS, FULL VEL
A HELP page	Copy the chapter, or all of HELP

HELP AND SEARCHING IT

HELP has this manual's text. Just type to search it: every match is highlighted and the chapter list shows only the chapters with matches. ENTER (or CMD G) goes to the next match, SHIFT ENTER the one before. ESC clears the search, then closes HELP.

In the dark skin, the plug-in pages (an effect opened from the mixer, a plug-in instrument's page) and the parameter pages are dark too.

WRITING PATTERNS

Each track has a note, an instrument number and an effect command per line: `C-2 01 956` means note C-2, instrument 01, command `9` with value `56`. A second effect column appears when you switch `FX2` on.

NOTES AND THE CURSOR

- With `• EDIT` on, the bottom rows of keys play and enter notes from the octave in `OCT`; the `Q` row is the octave above. `1` enters a note-off (`OFF`).
- In the instrument and command columns, type hex digits `0`–`9` `A`–`F`, or a letter `G`–`Z` for the extra commands.
- `SHIFT` `3` in a command column types `#`, the chord command (see Chords).
- `DELETE` clears the field under the cursor. `SHIFT` `DELETE` removes the line and pulls the track up; `ALT` `DELETE` inserts a blank line.
- A MIDI keyboard plays the current instrument and enters notes when EDIT is on. MIDI 60 = `C-3`.

RECORDING AND NOTE-OFFS

With EDIT on (`ESC` or `\`), playing while the song or block plays records the notes where you play them. On a plug-in or MIDI-out instrument, letting go of a key writes an `OFF` on the line you let go, so the notes are as long as you held them. When stopped (step entry), note-offs go in only if you ask: right-click `EDIT` and turn on `STEP ENTRY NOTE-OFFS`.

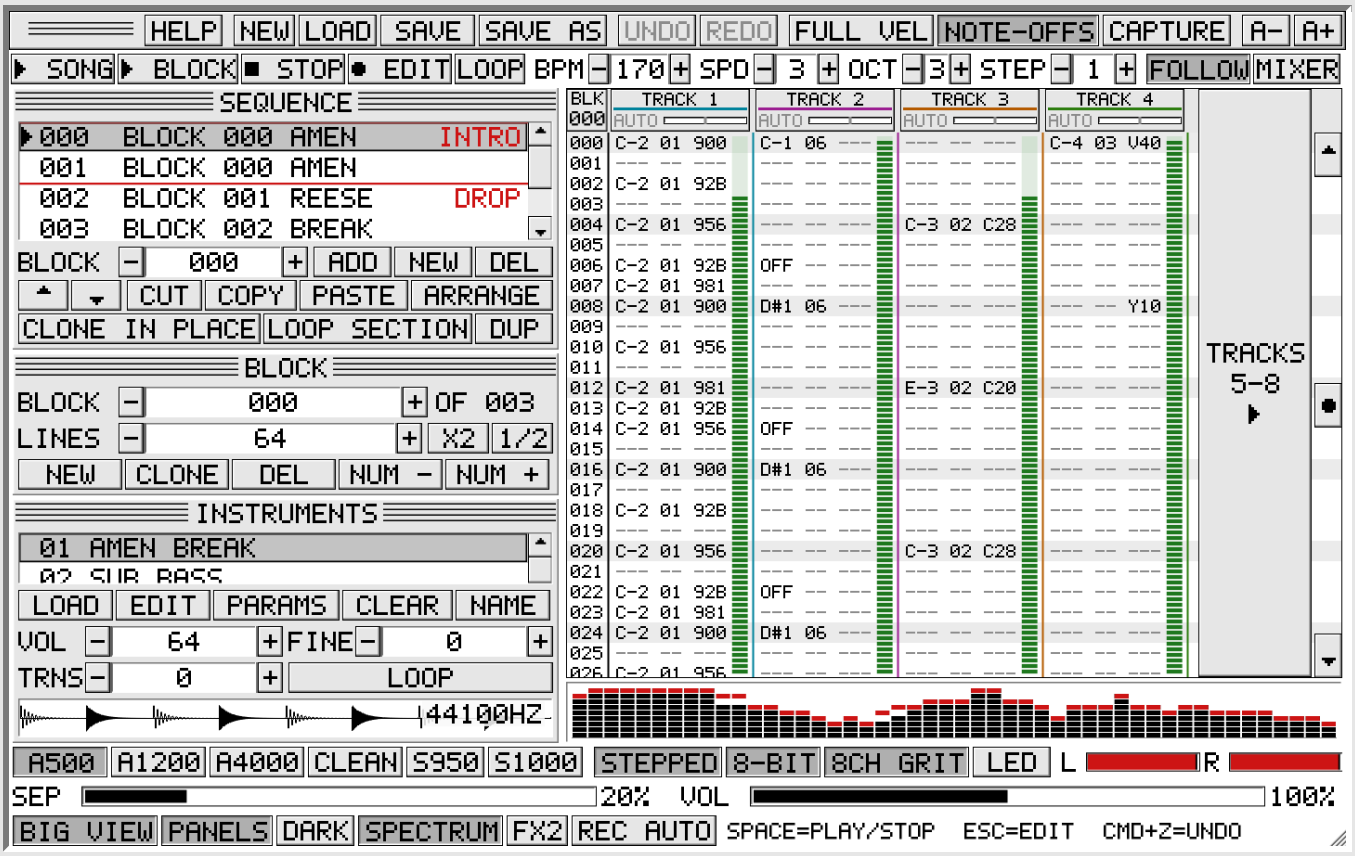
- **Legato:** let go of a key as (or just after) the next one goes down and no `OFF` is written – the next note ends the old one by itself. The OFF never lands on the next note, and the next note isn't pushed onto another track. CAPTURE does the same.
- `NOTE-OFFS` in the title bar (also right-click EDIT, and **Record Note-Offs** in the View menu) switches note-off recording off altogether: live, step entry and CAPTURE then write notes only.

- FULL VEL in the title bar (also right-click EDIT, and **Full Velocity** in the View menu) ignores how hard you hit a MIDI keyboard: every note plays and records at full velocity (127).

Both switches are remembered.

FOLLOW AND THE ARROWS

With FOLLOW on, the cursor moves with playback but the view doesn't scroll by itself: the play line runs down the screen and can go off the bottom. Scroll with the wheel or the arrows at the right of the pattern: ▲ / ▼ move a page (hold to keep going), • jumps to the play line. A new block starts at its top. When stopped, the arrows move the cursor a page.



PLAYING ON PAST THE BOTTOM: THE VIEW STAYS WHERE YOU LEFT IT

CAPTURE: KEEP WHAT YOU JUST PLAYED

The app is always listening to the MIDI keyboard and the laptop keys, even with EDIT off - the last two minutes. Noodle away; if you like what you played, press CAPTURE in the title bar or CMD R (or CAPTURE WHAT I PLAYED in the View menu, or right-click EDIT) and it is written into the pattern, one undo step.

- Played while the song or block was playing: the notes go where they were played, on the nearest line, with an `Mxx` for notes more than a third of a tick early or late.
- **One take is one time round.** Each time playback goes back to the start (the end of the song, a looping block, the loop lines) a new take begins, so loops never pile up. CAPTURE takes the latest time round that had notes in it: play on the 2nd time round, let it go round 5 more times, and it still writes the 2nd. A note held over the wrap gets its `OFF` where you let go, early in the block, so it rings over the loop as played.
- Played while stopped: the notes since the last pause of more than 4 seconds go in from the cursor, at the song's tempo.
- Notes start on the cursor's track; held notes spread over the tracks to the right. Chords that match the chord table go in as one note + `#xy` (see Playing chords in). Plug-in and MIDI-out instruments get note-offs (not between legato notes, and none with NOTE-OFFS off).

HIGHLIGHTING WITH THE MOUSE

Drag across the pattern to highlight notes and effects; it snaps to whole fields and can cover several tracks. `SHIFT`-click highlights from the cursor to the click, and `CMD` `A` highlights the whole block.

- `CMD` `C` / `CMD` `X` copy or cut the highlight; `CMD` `V` pastes at the cursor, each part into the same kind of field. Without a highlight they work on the whole track (add `SHIFT` for the block).
- `CMD` `↑` / `CMD` `↓` transpose the track a semitone, with `SHIFT` an octave.
- Right-click inside a highlight to bounce it to a new sample.

LOOPING SOME LINES

Drag down the line numbers to mark lines; they turn red and playback goes round them. `CMD` `[` and `CMD` `]` set the start and end at the cursor. `LOOP` on the transport row turns the loop off and on again.

PLAYING A BLOCK

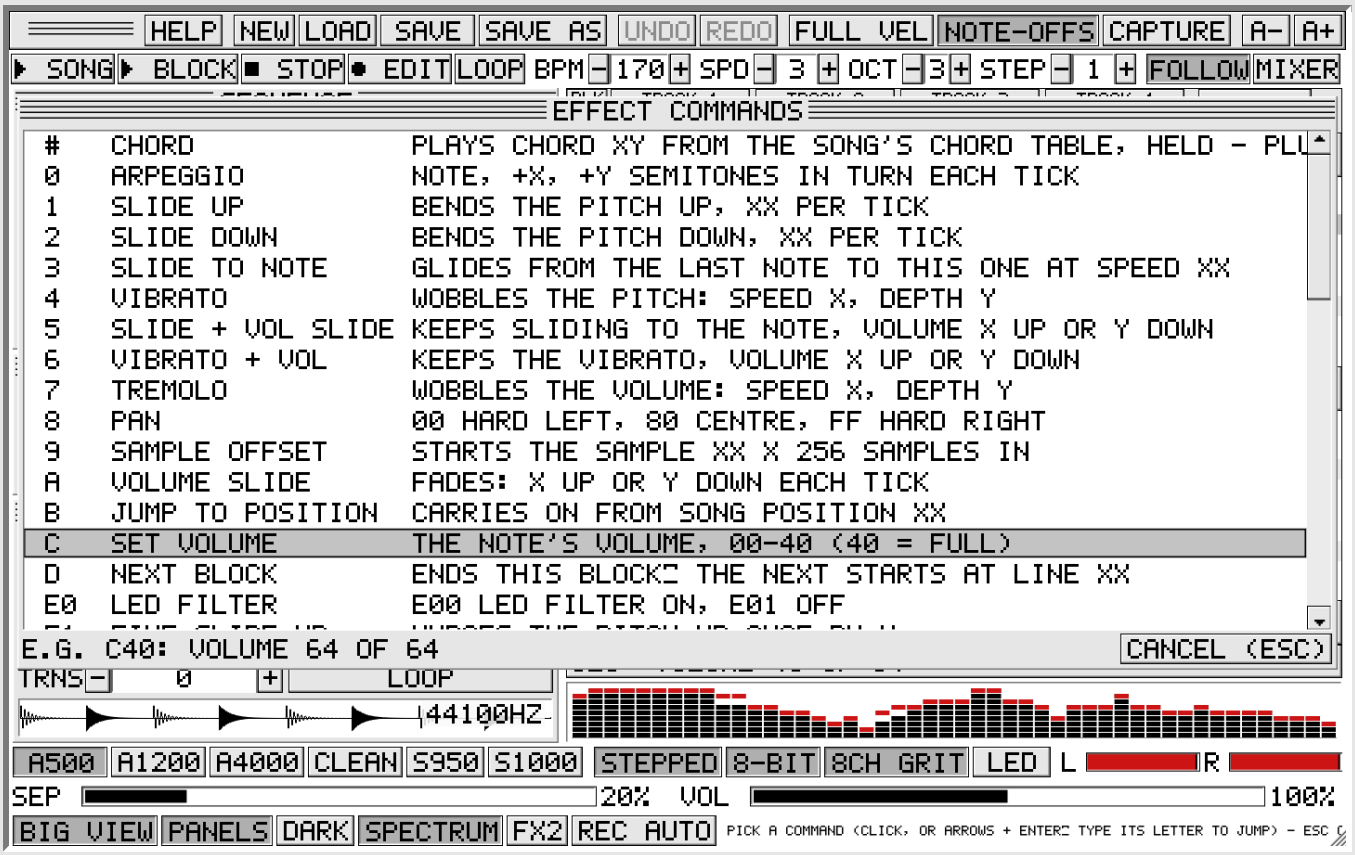
While a block plays, picking another block (in the sequence, the BLOCK box or the BLK corner) carries playback over to it on the same line, without stopping, so you can try blocks against each other in time.

UNDO

One history covers everything: notes, blocks, the sequence, instruments, samples, the mixer and song settings. CMD Z and CMD SHIFT Z work on every page, and the status line says what was undone. It keeps the last 200 steps.

EFFECT COMMANDS

Right-click an effect cell for a list of every command with what it does. Click one, or type its letter, and it goes into the cell with a useful starting value, the cursor ready on the value. Put the cursor on any command to see it explained in the help strip under the pattern.



RIGHT-CLICK AN EFFECT CELL: THE COMMAND LIST

THE CHORD COMMAND (#)

`#xy` plays chord slot `xy` from the song's chord table, held, on a plug-in or MIDI-out instrument. It is at the top of the command list; clicking it opens the chord list. See Chords.

CLASSIC COMMANDS (0–F)

The same numbering as ProTracker, so MOD files load straight in.

CMD		EFFECT
`0xy`		Arpeggio (+x, +y semitones), as in ProTracker
`1xx`	`2xx`	Slide pitch up / down

CMD	EFFECT
3xx	Slide to the note
4xy	Vibrato (speed x, depth y)
5xy 6xy	Slide to note / vibrato, plus volume slide
7xy	Tremolo
8xx	Pan: 00 left, 80 centre, FF right
9xx	Sample offset (xx × 256)
Axy	Volume slide (x up, y down)
Bxx	Jump to song position xx
Cxx	Volume (00 - 40); velocity on plug-in and MIDI-out instruments
Dxx	Next block, starting at line xx
E0x	LED filter: E00 on, E01 off
E1x E2x	Fine slide up / down
E4x E7x	Vibrato / tremolo waveform
E5x	Finetune
E6x	Loop: E60 marks the start, E6x repeats x times
E8x	Coarse pan
E9x	Retrigger every x ticks
EAx EBx	Fine volume up / down
ECx	Cut the note at tick x
EDx	Delay the note by x ticks
EEx	Delay the line by x lines
Fxx	01 - 1F speed, 20 - FF BPM, 00 stop

EXTRA COMMANDS (G–Z)

Type the letter in an effect column, then two hex digits.

CMD	EFFECT
Gxx	Glide to the note over xx ticks

CMD	EFFECT

`Hxx`
`Lxx`
`Wxx`
	High-pass / low-pass / band-pass filter, cutoff xx (`00` 20 Hz – `FF` 20 kHz). On a plug-in or MIDI-out instrument, its learned knobs instead.

`Qxx`	Filter resonance (or a learned knob)
`Ixx`	Transpose just this note: `80` none, `8C` +12, `74` -12
`Kxx`	Chance: the step plays xx/FF of the time
`Mxx`	Micro-move: `01`–`7F` late by xx/128 of a line, `81`–`FF` early
`Nxx`	Note length: release after xx ticks
`Oxx`	Overdrive (on plug-in / MIDI-out instruments: the mod wheel)
`Pxx`	Start at xx/256 of the sample
`Rxy`	Roll: retrigger every x ticks, y `1`–`7` quieter or `9`–`F` louder each time
`Sxx`	Play slice xx+1
`Txx`	Micro-tune: `80` = 0, each step 1 cent
`U01` `U00`	Play backwards / forwards
`Xxy`	Randomise: x `0` note ±y, `1` volume, `2` pan, `3` start
`Zxx`	Bit crush to xx bits
`Vxx` `Jxx`	The track's automation knobs AUTO A and AUTO B (see Automation)
`Yxx`	The next knob change on this track glides over xx ticks

BLOCKS & THE SONG

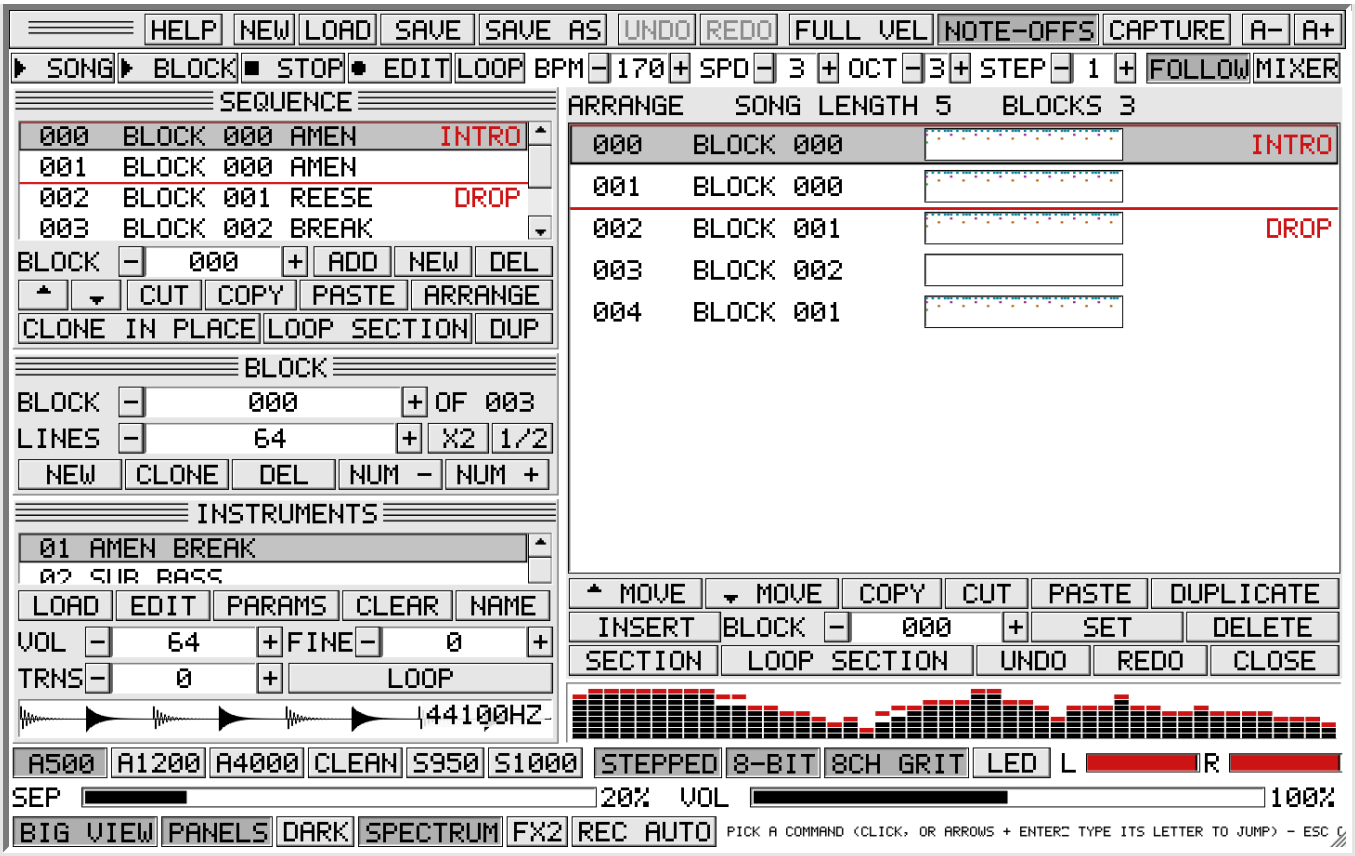
A song is a list of positions, each playing a block. The same block can play at many positions; change it once and every position hears the change.

THE SEQUENCE PANEL

- ADD puts a new position after the selected one, playing the same block; change it with BLOCK - +. NEW makes a new empty block in a new position.
- Drag over positions to highlight them, then CUT COPY PASTE DUP DEL. Drag the highlight, or use ▲ ▼, to move positions.
- CLONE IN PLACE gives a position its own copy of its block, in the next free block, so you can change it without touching the other places it plays.
- Click at the right of a row to start a named **section** there (INTRO, DROP…). LOOP SECTION loops the section you're in; pick a position in another section and playback goes there when the current block ends.
- **Block names:** double-click a position (or right-click → BLOCK NAME...) to name its block (AMEN, REESE…). The name shows after the block number, before any section name.

THE ARRANGE PAGE

ARRANGE shows the song order full size, with a picture of each block's notes. It shares the highlight and clipboard with the sequence panel. INSERT adds the BLOCK number after the cursor, SET makes the highlighted positions play it.



THE ARRANGE PAGE, WITH TWO SECTIONS

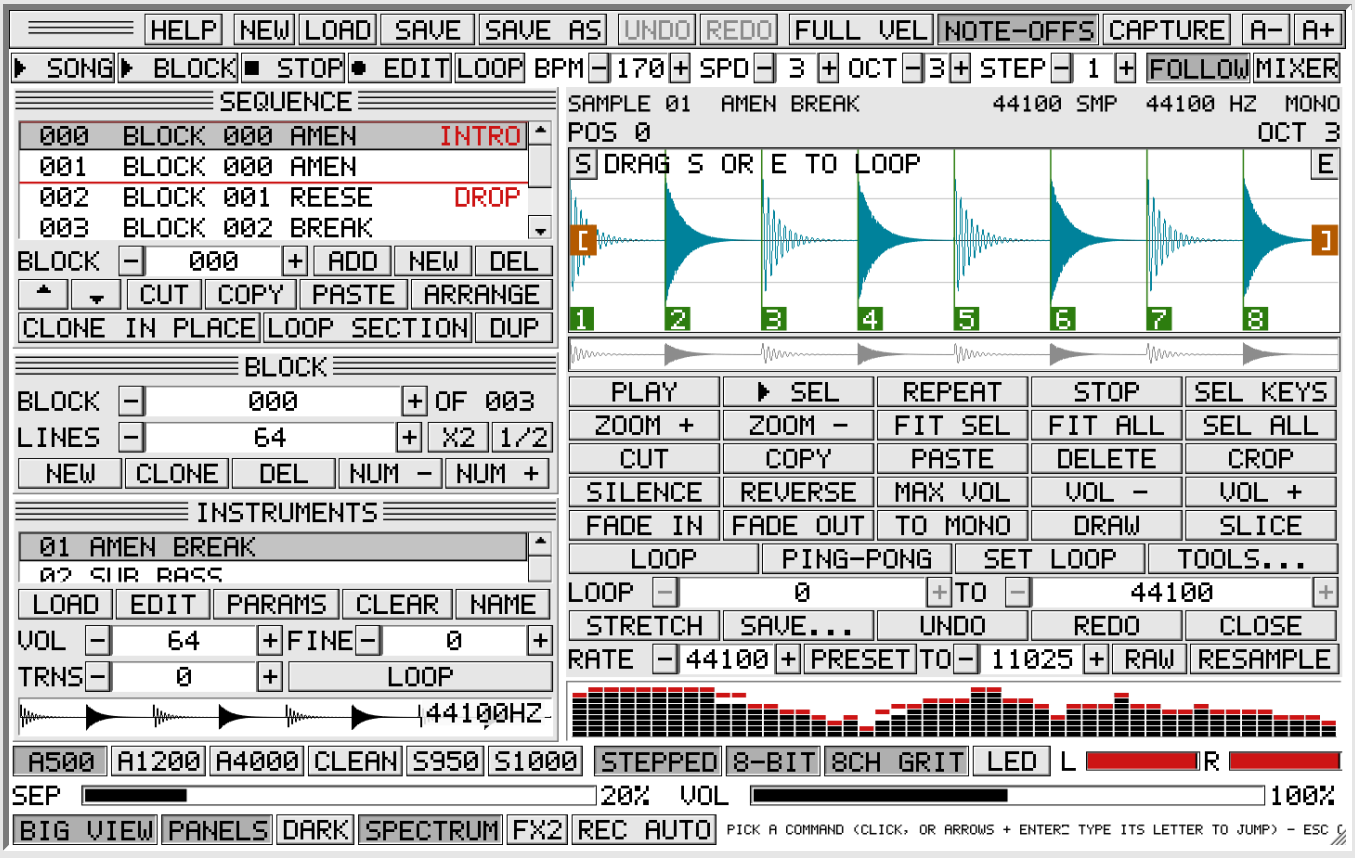
INSTRUMENTS & SAMPLES

THE SAMPLE BROWSER

LOAD opens the browser in place of the pattern. Every sample you highlight plays straight away, and the note keys or a MIDI keyboard play it at other pitches. Shortcut buttons go to Home, Music, Desktop, Downloads, Documents and Drives, and four FAV buttons hold your own folders (click an empty one to set it; SHIFT-click changes it, ALT-click clears it). ENTER or a double-click loads the sample.

THE SAMPLE EDITOR

EDIT (or a double-click on the little waveform) opens the editor.



THE SAMPLE EDITOR, WITH SLICES

- Drag to select; grab either end of a selection to move just that end. The note keys and MIDI play the sample through the full instrument sound; SEL KEYS plays just the selection.
- CUT COPY PASTE DELETE CROP SILENCE REVERSE MAX VOL VOL - VOL + FADE IN FADE OUT TO MONO, and DRAW to draw the wave.

- TOOLS...: GAIN, NORMALISE, REMOVE DC, TRIM SILENCE, INSERT SILENCE, CROSSFADE LOOP (a smooth jump back), PITCH (semitones, resampled), SWAP LEFT / RIGHT, START / END from the selection, SAVE SAMPLE. SAVE... writes the sample to a WAV wherever you choose.
- **START / END:** drag the green [and] handles at the middle of the wave: only the part between them plays, the rest is shaded but kept. They're separate from the loop (which is cut at END); 9xx counts from START.
- SPACE stops whatever plays, the song too; when nothing does, it plays the sample.
- **Loop:** drag the S and E handles at the top of the wave; you hear the loop while you drag. PING-PONG plays it forwards then backwards.

RATES

Each instrument has a RATE: the speed it plays at on C-3. Samples load at their own rate, so they sound as recorded. RATE changes it without touching the data (lower rate, lower and slower); RESAMPLE converts the data to a new rate and keeps the pitch, with RAW for the rough, unfiltered old-sampler way.

SLICING

SLICE opens the slicer. AUTO finds the hits (SENS sets how quiet), EVEN cuts equal slices. Drag a tab to move a slice point, double-click the wave to add one, ALT-click a tab to remove it. With SLICE TO KEYS on, C-3 plays slice 1, C#3 slice 2 and so on; without it, Sxx picks the slice.

TIME STRETCH

STRETCH opens the old-sampler stretch: **CLASSIC** (the Akai sound, matched to recordings of the real thing), **REVISED** (cleaner joins), **HARD** (no blending) and **AKAIZER** (Akaizer v1.2's granular method, rebuilt). Set TIME %, SEMI and CENTS and the CYCLE length; FIT TO LINES makes the selection last a number of lines at the song's tempo. PREVIEW, then APPLY.

SAMPLING: RECORDING AUDIO

Record straight into the selected instrument, from an audio interface, the mic or line-in, or from the app's own output. Open the SAMPLE page with CMD 4, **Sample Audio…** in the View menu, right-click an instrument → **SAMPLE (RECORD AUDIO)…**, or click an empty instrument's **NO SAMPLE - CLICK TO RECORD**.



SAMPLING THROUGH THE S950, ITS INPUT DRIVEN INTO THE CLIP LIGHTS

CONTROL	WHAT IT DOES
INPUT RESAMPLE	Record the audio input (chosen in **AUDIO/MIDI SETTINGS…**, bottom right), or everything the app plays: play the song or a block, or jam, and record what comes out
MONITOR	Hear the input through the app. Off to start with: with a mic, use headphones or it feeds back
STEREO MONO LEFT RIGHT	What goes into the instrument (MONO mixes left and right)
MANUAL	Records from the moment you press RECORD
THRESHOLD	RECORD arms it; it starts when the sound goes over the THRESHOLD level (the red line on the meter), keeping the 20 ms before so the attack isn't cut

CONTROL	WHAT IT DOES
WITH PLAYBACK	RECORD arms it; it records from when the song or block starts to when it stops: one block, exactly
INPUT GAIN	-24 to +24 dB. The meter shows the level, a peak hold and a CLIP light
RECORD STOP	Start (or arm) and end it (R). Up to 5 minutes. It carries on if you go to another page
KEEP AGAIN	The take into the instrument, one undo step (ENTER), or thrown away and recorded again (BACKSPACE)
TRIM SILENCE NORMALISE	Cut quiet at both ends; bring the loudest point to full level. Both on to start with
CLEAN AMIGA S950	How the sound gets in. CLEAN (the default): the interface as it is. AMIGA: 8-bit at 8363 / 16726 / 22050 / 28867 Hz with no anti-alias filter (the aliasing grit). S950: 12-bit at 7.5K–48K through its filter, which follows the rate. You hear it as you sample, with MONITOR on
8 BIT 12 BIT	The bits, for either sampler (an S950 at 8 bits, an Amiga at 12)
INPUT DRIVE	0 to +24 dB into the sampler's input: the S950 goes soft and fat then hits the ceiling (the CLIP lights), like the classic overloaded records; an Amiga sampler just clips
AMIGA RATE	CLEAN only: NONE keeps the recording's rate; 8363, 16726, 22050 or 28867 Hz resamples it as you keep it, at the same pitch

The first time on a Mac, macOS asks whether Paula's Revenge may use the microphone: say OK, or the input is silent (System Settings → Privacy & Security → Microphone changes it later). After KEEP it's a sample like any other: EDIT to slice or loop it, and it's saved in the song.

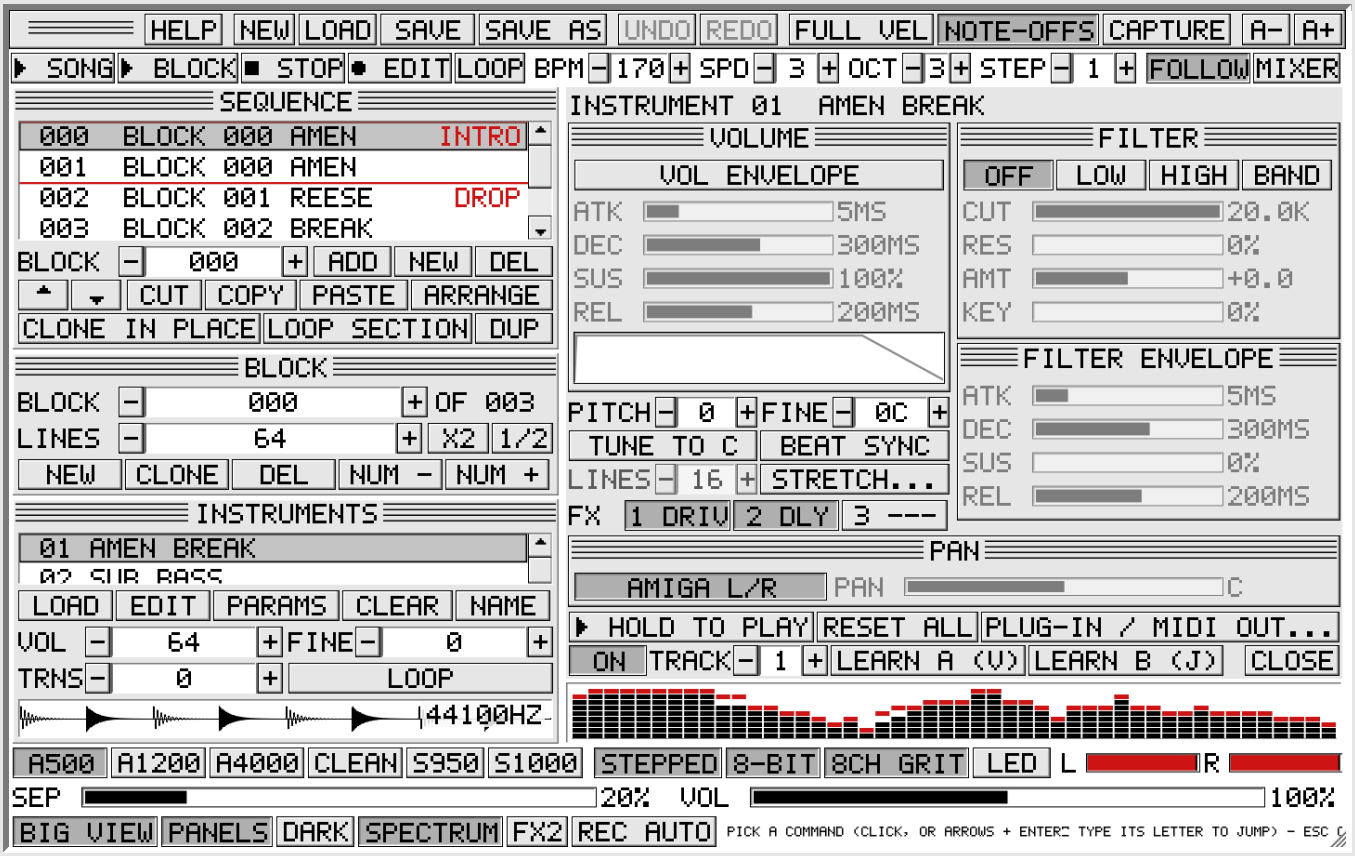
INSTRUMENT VOL AND SWITCHING OFF

VOL is a level for the whole instrument on top of each note's Cxx. Any instrument can be switched off with its ON button (on its

PARAMS page): its notes are silent and the list shows it dimmed with `OFF`.

SOUND SHAPING

PARAMS opens the current instrument's page. Everything starts off or neutral, so MODs and older songs sound exactly as they did.



PARAMS: ENVELOPE, FILTER, PAN, AND THE ON BUTTON

- **Volume envelope:** VOL ENVELOPE on, then ATK, DEC, SUS, REL (1 ms – 10 s). With it on, a note-off starts the release instead of cutting the note.
- **Filter:** OFF / LOW / HIGH / BAND, CUT, RES, AMT (how far the filter envelope moves it) and KEY (how far the cutoff follows the note). The filter envelope has its own ATK, DEC, SUS, REL.
- **Pan:** AMIGA L/R follows the Amiga track layout; switch it off to give the instrument its own PAN.
- **Tuning:** PITCH (±48 semitones) and FINE (±100 cents). TUNE TO C listens to the sample and sets them so C-3 plays a C.
- **BEAT SYNC** (like Renoise): the sample lasts LINES lines at C-3 at any tempo, repitched live to fit. STRETCH... opens the time stretch set to those lines, to fit it at its own pitch instead.

- ▶ HOLD TO PLAY or the keys to hear it; RESET ALL puts everything back; double-click any slider for its default.
- ON switches the instrument off and on. To switch it from the pattern, pick a track, press LEARN A (V) or LEARN B (J), then click ON: that track's V00 switches it off and V01–VFF on.
- PLUG-IN / MIDI OUT... plays the instrument with a VST3 / AU instrument or with your hardware instead (see Plug-ins and MIDI out).

INSTRUMENT EFFECTS

Every instrument has THREE INSERT SLOTS OF ITS OWN, like a mixer strip's: the built-in filter, drive, compressor, delay, reverb and chorus, or a VST3 / AU plug-in. They're on the FX row, under the volume envelope in the picture above (and on a plug-in instrument's page).

- Click a slot to open it on the mixer's effect page; BACK returns to the instrument. Right-click for OPEN, BYPASS, COPY, PASTE and CLEAR. A lit slot is on.
- The sound goes through the instrument's effects first, then into the strip of the track it plays on. Every note goes through the one chain, so a delay or reverb carries on across notes; its output goes to the track of the instrument's **latest** note.
- The effects belong to the instrument: saved in the song, undo steps, and copied with it (COPY / PASTE / DUPLICATE). RENDER WAV and bouncing include them. A plug-in's own delay on an instrument slot isn't compensated.

PAULA & THE AKAIS

The output strip picks the machine you hear. None of these change your samples; switch between them any time.

CONTROL	WHAT IT DOES
A500	The A500 (and A600, A2000): a 4.42 kHz output filter, warm and dull
A1200	The A1200: much brighter, about -3 dB around 14 kHz, as measured
A4000	The A4000: about 4.6 kHz, as measured
CLEAN	Paula with no output filtering
S950	Akai S950: 12-bit, stepped variable-rate playback, its 6-pole filter
S1000	Akai S1000: 16-bit, interpolated playback, clean 44.1 kHz-style output
STEPPED	Paula's stepped playback, exactly as the chip makes it (off = smooth)
8-BIT	Samples play at 8-bit resolution
8CH GRIT	OctaMED's 8-channel sound on all 8 tracks. The Amiga has 4 channels, so 8-channel trackers mixed two tracks into each (1+5, 2+6, 3+7, 4+8) at half level and squashed them back to 8 bits; this does the squashing, then brings the level back up. On for the Amiga models. Every track plays at full level either way, and a safety limiter stops eight loud tracks clipping
LED	The Amiga's "LED" low-pass filter (12 dB/oct at 3.2 kHz) at the start of the song (`E00` / `E01` switch it while playing)
SEP	How far tracks lean to the Amiga sides (1, 4, 5, 8 left; 2, 3, 6, 7 right). 20% to start; 0% = all centred

HOW THE PAULA MODES WORK

- THE STAIRCASE, WITHOUT FALSE NOTES. Paula holds each sample until the next, so its output is a staircase whose copies of the sound go far above hearing. Played straight, those copies fold back as out-of-tune tones. With STEPPED on, every change of level is a band-limited step instead: everything a real Paula puts out below 20 kHz is there, its own staircase images included, and nothing folds back. It sounds the same at 44.1, 48 or 96 kHz.

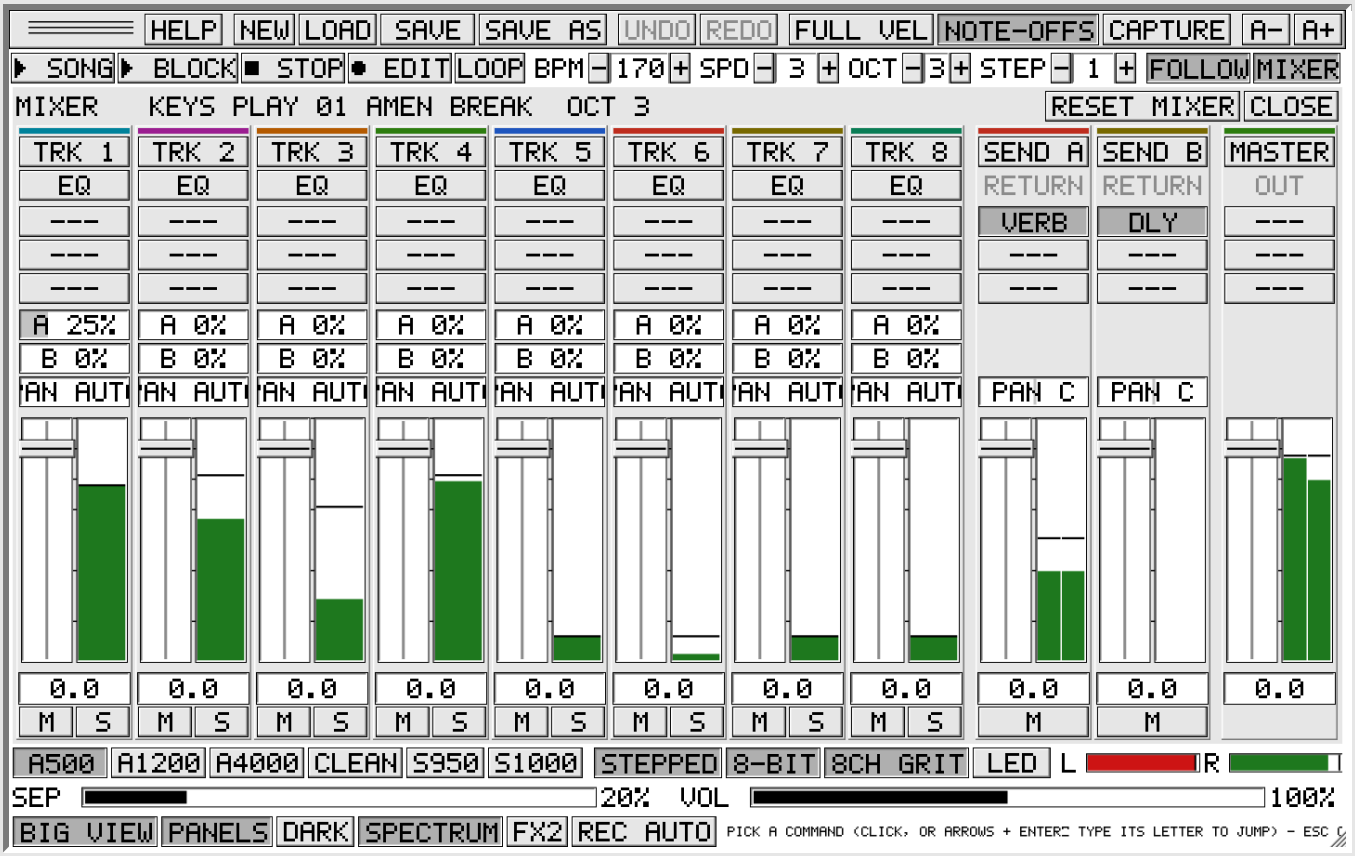
- VOLUME BELOW 64. At 64 a real Paula changes level exactly when the next sample is due; below 64 it works on a fixed 55.4 kHz grid, so each change waits for the next grid point. That adds a faint grit to everything not at full volume. A500, A1200 and A4000 do this; CLEAN doesn't.
- THE FILTERS are the real ones: the A500's 4.42 kHz RC filter (100 nF, 360 Ω), the A1200's and A4000's measured responses, the LED filter as a Butterworth. (From Henryk Richter's *Amiga Paula vs. System Theory*; the band-limited steps follow OpenMPT and Antti S. Lankila.)

PANNING, IN ORDER

1. `8xx` in the pattern (an absolute position, not scaled by SEP).
2. The track's pan, the bar under its name (AUTO = none).
3. The instrument's PAN, when its AMIGA L/R is off.
4. Otherwise Amiga style, by SEP. Stereo samples play centred with their own stereo image.

THE MIXER

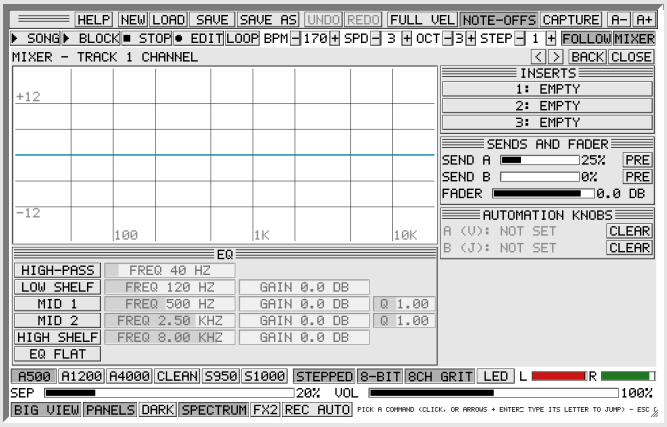
MIXER on the transport row (CMD 2) opens the mixer across the window; ESC or CMD 1 goes back. The keys still play the current instrument while it's open.



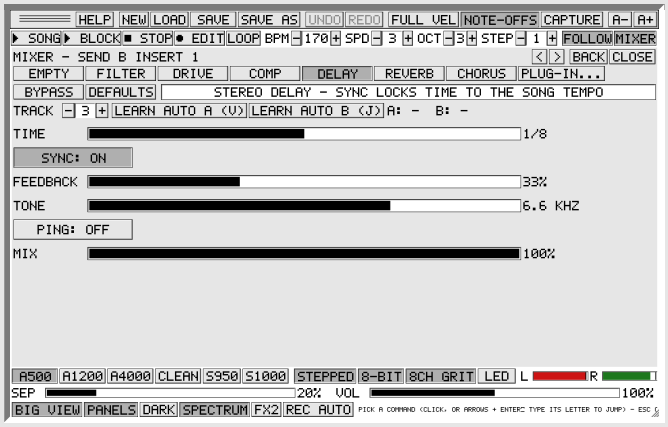
THE STRIPS: TRACKS, SEND A, SEND B, MASTER

STRIP PART	WHAT IT DOES
Name / EQ	Opens the track's channel page. EQ is lit when a band is on; EQ AB when automation knobs are set.
3 insert slots	Click for the effect page; SHIFT-click bypasses (a bypassed slot is dimmed).
A, B	How much goes to SEND A / B. AP / BP = pre-fader. Double-click = off.
PAN	The track's pan. Double-click = AUTO.
Fader	-INF to +6 dB with the meter beside it. SHIFT = fine, double-click = 0 dB.
M / S	Mute and solo.

EQ → INSERTS → PRE SENDS → FADER → POST SENDS → MASTER → PAULA / AKAI OUTPUT



CHANNEL PAGE: EQ, INSERTS, SENDS



EFFECT PAGE: SEND B'S DELAY

- **Channel page:** five EQ bands (HIGH-PASS, LOW SHELF, MID 1, MID 2, HIGH SHELF) with a response curve; click a band's name to switch it on. `<` `>` step through the tracks.
- **Effect page:** FILTER, DRIVE, COMP, DELAY (in ms or synced to the song), REVERB, CHORUS, or `PLUG-IN...`. Every page has `BYPASS` and `DEFAULTS`.
- **Returns:** SEND A starts with a reverb and SEND B with a delay, fully wet, silent until a track sends to them.
- **Delay compensation:** plug-ins that give their sound back late (lookahead limiters, linear-phase EQs) are lined up, so every track stays in time, sends included. The strips page title shows the total delay.

A mixer with nothing changed is skipped, so the song sounds exactly as it would without one. Everything in it is saved in the song, undoable and in renders.

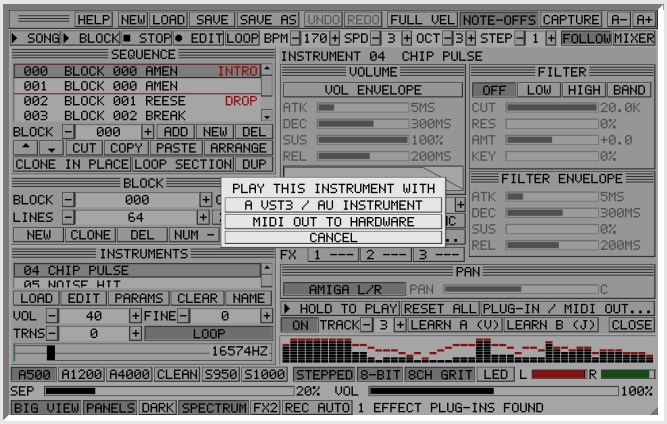
PLUG-INS

EFFECTS IN THE MIXER

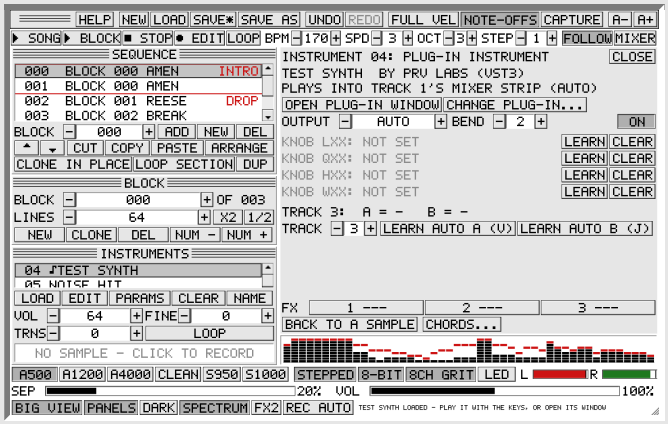
On any insert slot's effect page press `PLUG-IN...`. The first time, it looks for the plug-ins on your Mac, each one in a separate hidden copy of the app, so one that crashes can't take Paula's Revenge down. Double-click one to load it; `OPEN PLUG-IN WINDOW` shows its own controls. Its settings are saved in the song. If a song uses a plug-in this Mac doesn't have, the slot shows `!NAME`, passes the audio through and keeps the settings.

INSTRUMENTS

Pick an instrument, press `PARAMS`, then `PLUG-IN / MIDI OUT...` and `A VST3 / AU INSTRUMENT`.



PLUG-IN OR MIDI OUT?



A PLUG-IN INSTRUMENT'S PAGE

- `EDIT` opens the plug-in's own window. The laptop keys keep playing it while that window is in front.
- `OUTPUT` picks the mixer strip it plays into; AUTO is the track that played its latest note (the same in the song, a render and a bounce). `BEND` should match the plug-in's own pitch bend range, so slides and vibrato land in tune.
- Notes play one at a time per track. `Cxx` is velocity, pitch commands become pitch bend, `Oxx` is the mod wheel, and `ECx` `EDx` `Nxx` `Kxx` `Mxx` `E9x` `Rxy` work. Sample-only commands are ignored.
- Chords: `#xy` on a note's line plays chord slot `xy` from the song's chord table, held. See Chords.

- Letting go of a key while recording writes a note-off (see Recording and note-offs).
- LEARN next to L Q H W, then move a knob in the plug-in's window: those commands then move that knob on this instrument's notes.

AUTOMATION

Every track has two automation knobs, **AUTO A** (`Vxx`) and **AUTO B** (`Jxx`). Point one at any effect control and the pattern moves it.

1. Open an insert's effect page in the mixer. Pick the track in the `TRACK` box.
2. Press `LEARN AUTO A (V)`, then move the control: drag a built-in effect's slider, or turn a knob in the plug-in's window.
3. In the pattern, `V00` is the control's lowest and `VFF` its highest. The value holds until the next one.
4. Put `Yxx` before or beside the `V` to glide there over xx ticks instead of jumping.

- **On and off:** press LEARN, then click the effect's `BYPASS` (or an instrument's `ON`) instead. `V00` then switches it off and `V01`–`VFF` on.
- The channel page lists what each knob controls under AUTOMATION KNOBS, with CLEAR buttons. The help strip and the command list name it too.

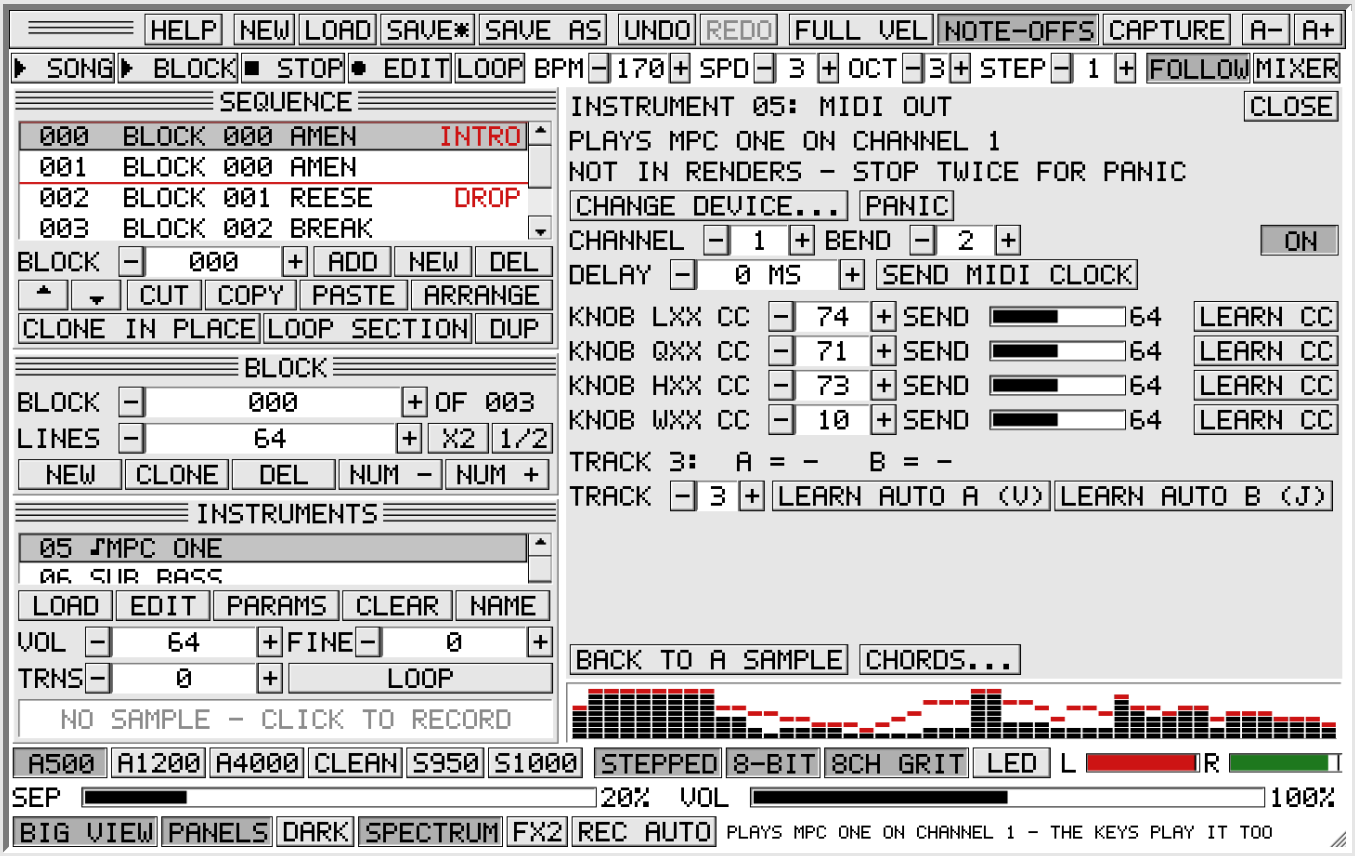
RECORDING BY HAND: REC AUTO

Switch on `REC AUTO` on the bottom row, turn EDIT on and play. Moving a learned control writes its command on the line that's playing: `V` or `J`, or `L` `Q` `H` `W` for an instrument's own knobs on the track playing it.

- It goes in the first effect column if that's free, otherwise in FX2. It never overwrites other commands; that line is skipped and the status line says so.
- One value per line per knob, and only when it changes. What the pattern plays back isn't recorded again.
- A whole pass, until you stop, is one undo step.

MIDI OUT

An instrument can play your hardware (an MPC, a microKORG, an NTS-1) over MIDI. Press PARAMS, PLUG-IN / MIDI OUT..., MIDI OUT TO HARDWARE, then double-click a MIDI output.



A MIDI-OUT INSTRUMENT'S PAGE

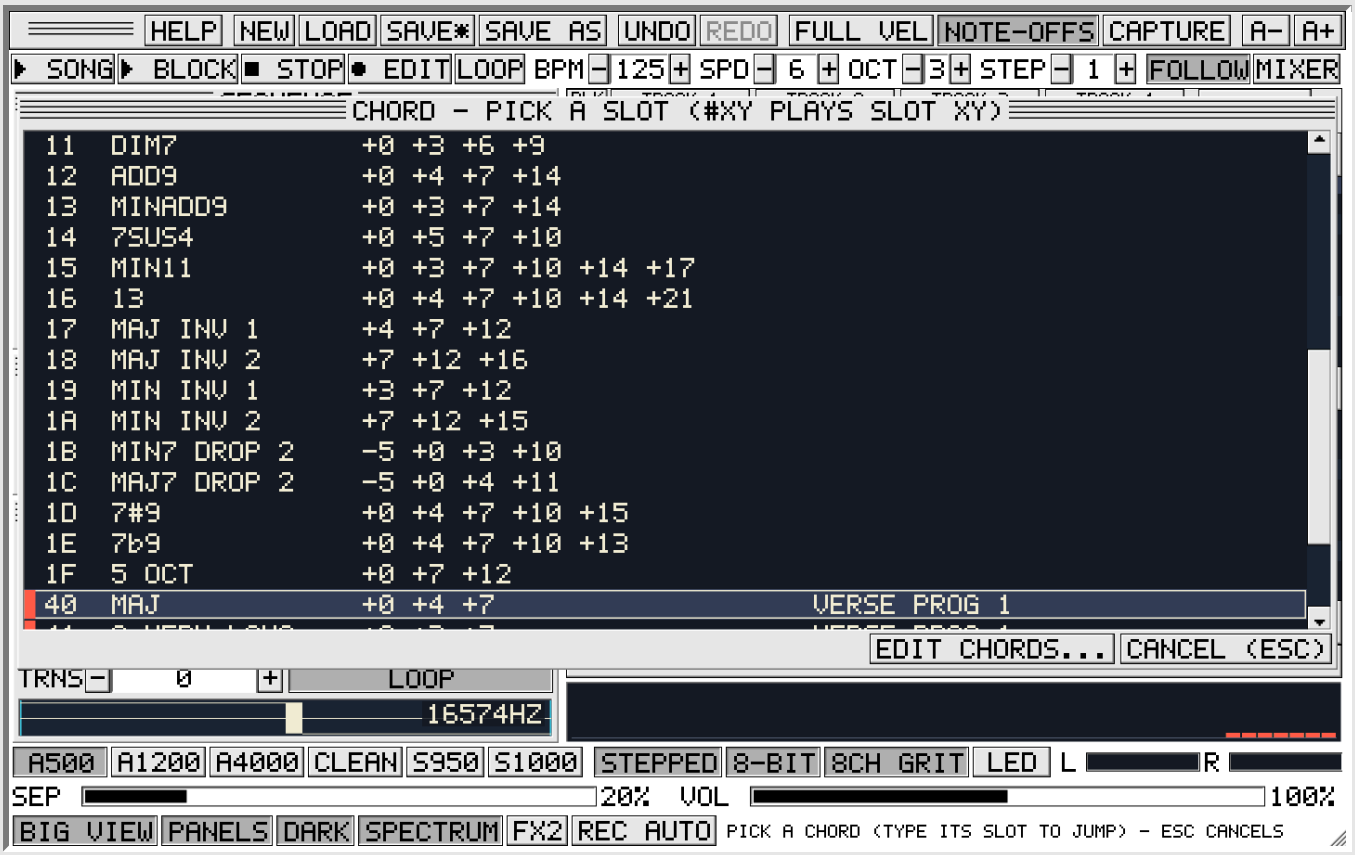
CONTROL	WHAT IT DOES
CHANNEL	MIDI channel 1-16
BEND	The hardware's pitch bend range, so pitch commands land in tune
ON	As for plug-in instruments; #xy plays chords here too
DELAY	-50 to +200 ms to line the hardware up with the audio. Positive: the MIDI goes later. Negative: all the audio goes later instead.
SEND MIDI CLOCK	24-a-beat clock with start, stop, continue and song position (4 lines = a beat), so the hardware follows the song

CONTROL	WHAT IT DOES
KNOB L Q H W	The CC each sends (74, 71, 73, 10 to start). LEARN CC takes the number from the next knob you turn on a controller; the SEND bar sends it now (and REC AUTO records it).
PANIC	All notes off on every MIDI output. STOP pressed while already stopped does the same.

Notes, velocity, pitch bend, the mod wheel and the timing commands work as for plug-in instruments. If the device isn't connected, the instrument shows !NAME, sends nothing and keeps its settings; CHANGE DEVICE... and the same device again reconnects it. MIDI-out instruments aren't in renders or bounces (they're hardware).

CHORDS

A plug-in or MIDI-out instrument plays a whole chord from one note. Put `#xy` on the note's line and it plays `CHORD SLOT XY` from the song's chord table, held. The note you type is the chord's root: `C-3` with `#01` (slot 01, MIN) plays C, D#, G.



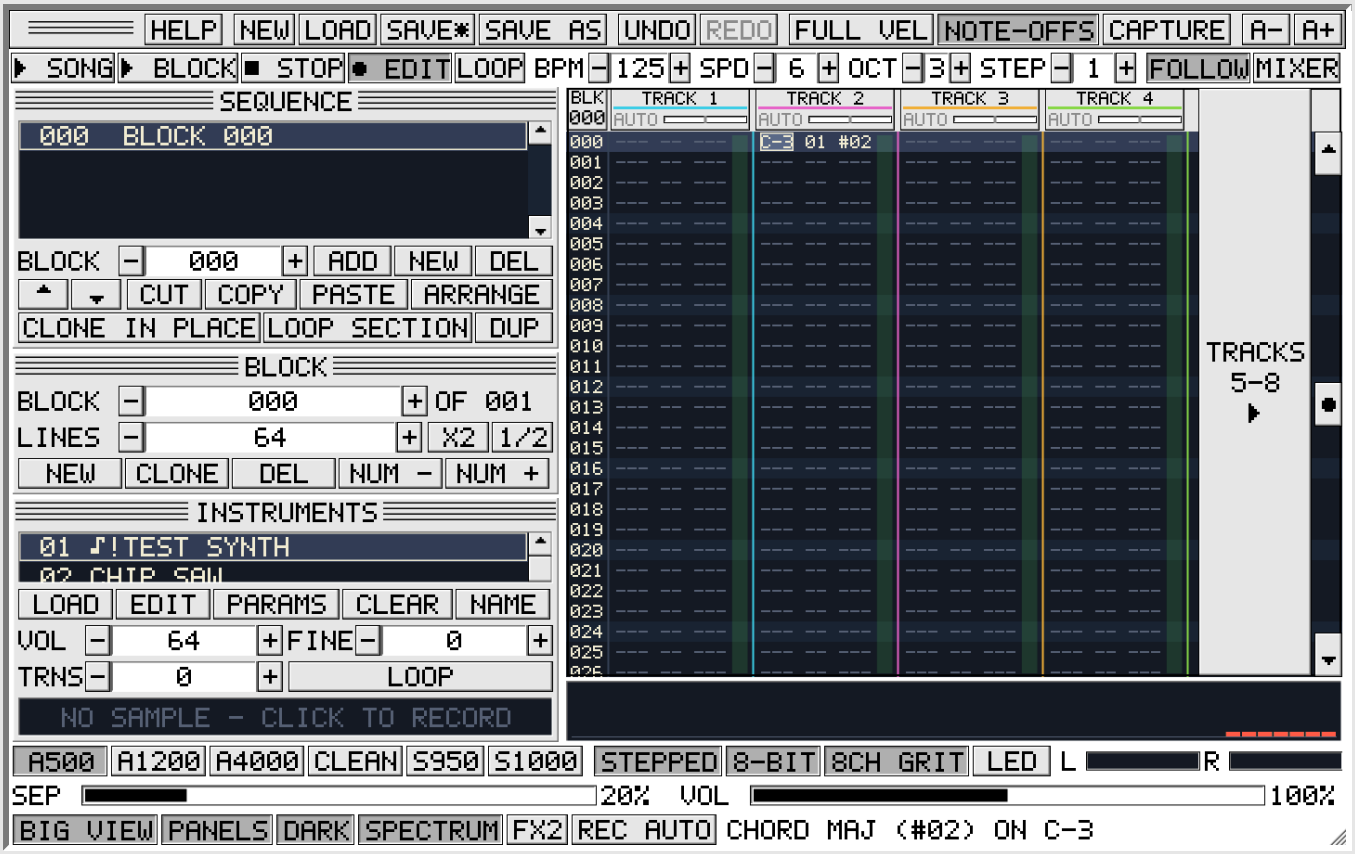
RIGHT-CLICK A `#` CELL: THE CHORD LIST

- Three ways in: type `SHIFT` `3` (`#`) in the command column, then the slot number; or right-click an effect cell and click `# CHORD` at the top of the list, then pick the chord; or right-click a cell that already has a `#` to change its chord.
- In the chord list, click a chord, or use the arrows and Enter; typing a slot number jumps to it. `EDIT CHORDS...` opens the CHORDS page on that slot, `EFFECT COMMANDS...` the usual list. Cancelling leaves `#01`.
- A slot holds up to 8 notes, below the root as well as above it. An inversion doesn't put the root at the bottom: `C-3` with MAJ INV 1 plays E, G and the C above.

- The chord ends like a note: the next note on the track, note-off, `ECx` or `Nxx`. Pitch commands bend the whole chord. On a sample instrument `#xy` plays just the note, and saving a MOD leaves it out.

PLAYING CHORDS IN

With EDIT on, or for CAPTURE, just play the chord on a MIDI keyboard or the laptop keys. On a plug-in or MIDI-out instrument, notes pressed together (within 60 ms) that match a slot in the song's chord table go in as `ONE NOTE, THE ROOT, WITH #XY` on one track. The status line says what it found, for example `CHORD MIN7 (#03) ON A-2`.



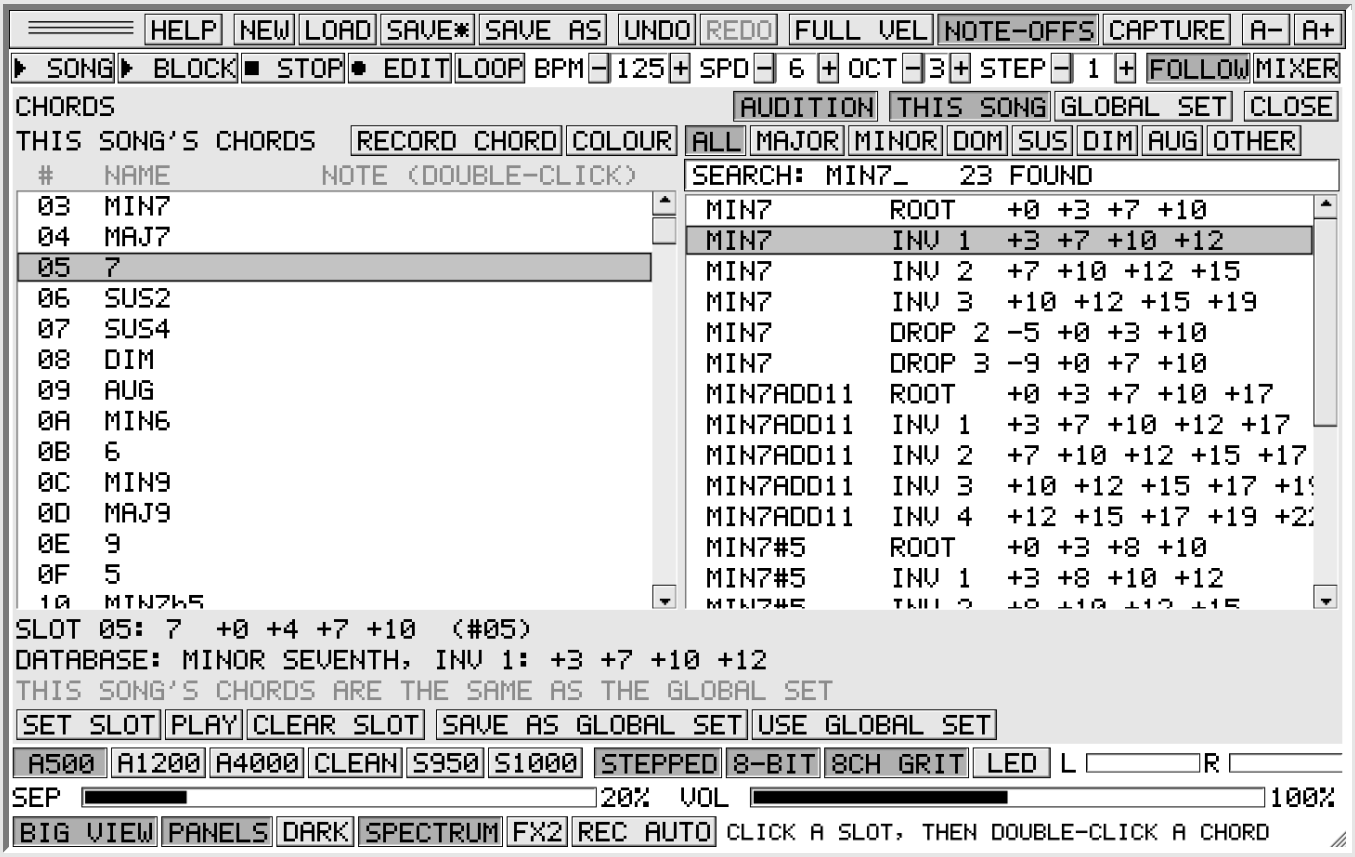
C, E AND G PLAYED TOGETHER: ONE NOTE, `#02` (MAJ)

- Inversions and spread voicings match their own slots (MAJ INV 1, MIN7 DROP 2…). If no slot has exactly the notes you played, a slot with the same note names in another octave is used, and the status line says the voicing may differ.
- When two slots have the same notes (C6 and Am7), the one whose root is your lowest note wins, then the one starting on the first key you pressed, then the lower slot.

- The notes go in as you play them; the moment they make a chord they're rewritten as one, still one undo step. In step entry the cursor moves on once for the chord. Its `OFF` goes on the line where you let go of the last key.
- `#` goes in the first effect column, or the second if the first already has a command. If both are taken, the notes go in one by one and the status line says so.
- Notes rolled slower than 60 ms stay separate notes, and sample instruments always get the notes as played. Right-click `EDIT` → `DETECT CHORDS` turns it off; it's remembered.

THE CHORDS PAGE

Press `CMD` `3`, or `CHORDS...` on the instrument's page. The slots 01–FF are on the left, the chord database on the right.

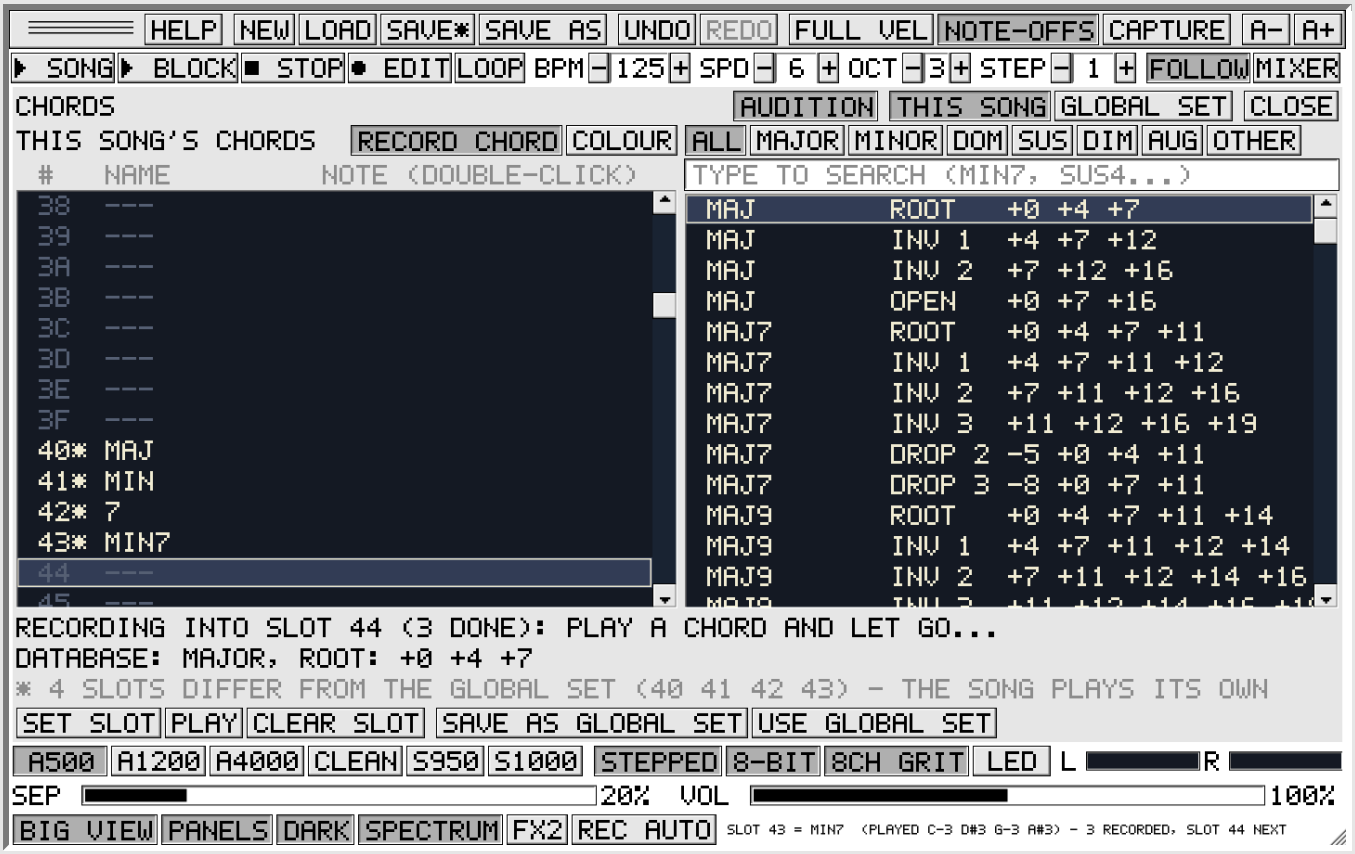


THE CHORDS PAGE, SEARCHING FOR MIN7

CONTROL	WHAT IT DOES
Slot list	Fixed columns: `#`, NAME (a long one ends in `..`) and NOTE, so the notes line up. Click a slot to fill it; drag, `SHIFT`-click or `CMD`-click to pick several. `SHIFT` `↑` `↓` moves through the slots. A `*` marks a slot that isn't the same in the global set; a coloured stripe, its group colour.
Double-click NOTE	Type the slot's note right there (Enter keeps it, Esc doesn't)
`RECORD CHORD`	Stays on: play chord after chord into the slots (see below)
`COLOUR`	A group colour for the picked slots: 8 colours or none
`AUDITION`	Clicking a chord plays it, on the last key you played
Database	Double-click a chord (or press Enter, or `SET SLOT`) to put it in the slot. `↑` `↓` move through it.
`ALL` `MAJOR` `MINOR` `DOM` `SUS` `DIM` `AUG` `OTHER`	Show one family of chords
Typing	Searches: `MIN7`, `SUS4`, `DROP 2`, `7B9` for 7♭9. Backspace and Esc clear it.
`PLAY`	Plays the chord with the current instrument, on the last key you played (Space)
`CLEAR SLOT`	Empties the slot, or all the picked ones (Delete): their `#xy` then plays just the note
`THIS SONG` `GLOBAL SET`	Which table you're editing
`SAVE AS GLOBAL SET`	The song's chords become the global set (it asks first)
`USE GLOBAL SET`	The song gets the global set (undo takes it back)

RECORDING YOUR OWN CHORDS

Click the slot to start at and press RECORD CHORD above the slot list: it stays down. Play chord after chord on a MIDI keyboard or the laptop keys, letting go between them. Each chord goes into the selected slot, then the selection moves on to the **next empty slot**, so nothing is written over; the status line counts them. Click RECORD CHORD again (or ESC) to stop.



THREE CHORDS RECORDED: 40, 41, THEN 43 (42 ALREADY HAD ONE); 44 IS NEXT

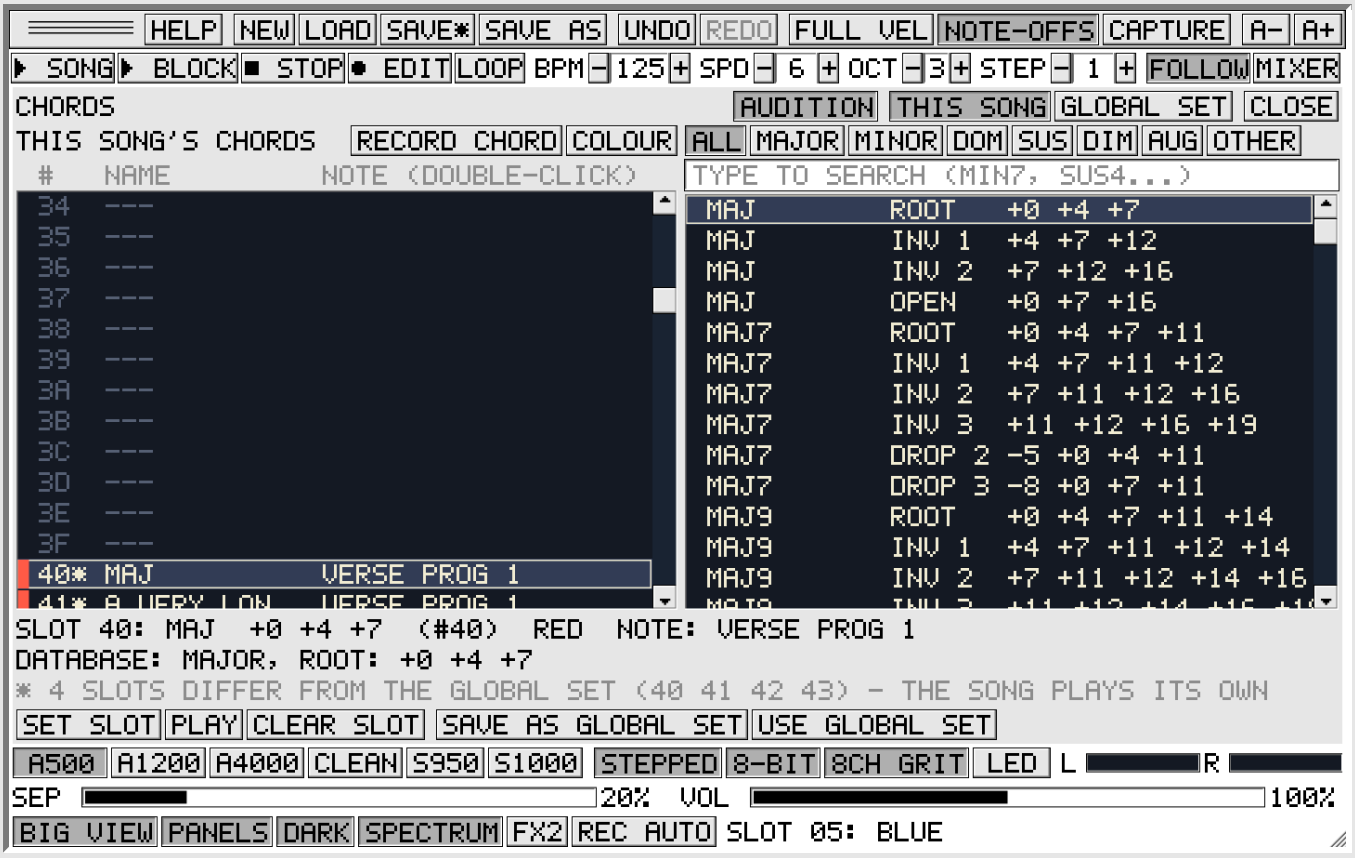
- If the notes are one of the database's chords, the slot takes that name and root: E G C becomes MAJ INV 1, so C-3 with it plays your E G C. Anything else is called USER and is measured from your lowest note.
- Up to 8 notes (with more, the lowest 8). One note on its own isn't a chord and is ignored.
- While it records, the MIDI keyboard doesn't write into the pattern and the laptop keys don't type into the search. Each chord is an undo step.

NOTES ON SLOTS

Double-click a slot's NOTE column (or right-click → NOTE...) and type a note of your own right there, such as `VERSE PROG 1`, up to 24 characters; Enter keeps it, Esc doesn't. It shows in the slot list's NOTE column, in the chord list over the pattern and in the help line for a `#xy`, and it stays with the slot when you change its chord. Notes are saved in the song and in the global set.

COLOUR GROUPS

Pick several slots (drag down the list, `SHIFT`-click a range, `CMD`-click one at a time), then `COLOUR` (or right-click → COLOUR...): red, orange, yellow, green, cyan, blue, purple, pink, or none. The colour shows as a stripe left of the slot number, in the chord list over the pattern, and on the `#xy` in the pattern itself, so a progression stands out while you write.



CHORD GROUPS IN THE PATTERN: `#40` RED, `#05` BLUE

COPY, PASTE and CLEAR work on all the picked slots. Colours are saved in the song and in the global set.

HEARING CHORDS (AUDITION)

With `AUDITION` on (top right of the CHORDS page; on to start with), clicking a slot or a chord in the database plays it with the current instrument while you hold the mouse, on **the last key you played** (C of the octave before you've played any). The arrow keys play each chord briefly as you move. In the chord list over the pattern, moving with the arrow keys or the mouse plays each chord on that cell's note with that track's instrument; clicking still picks it.

THE DATABASE

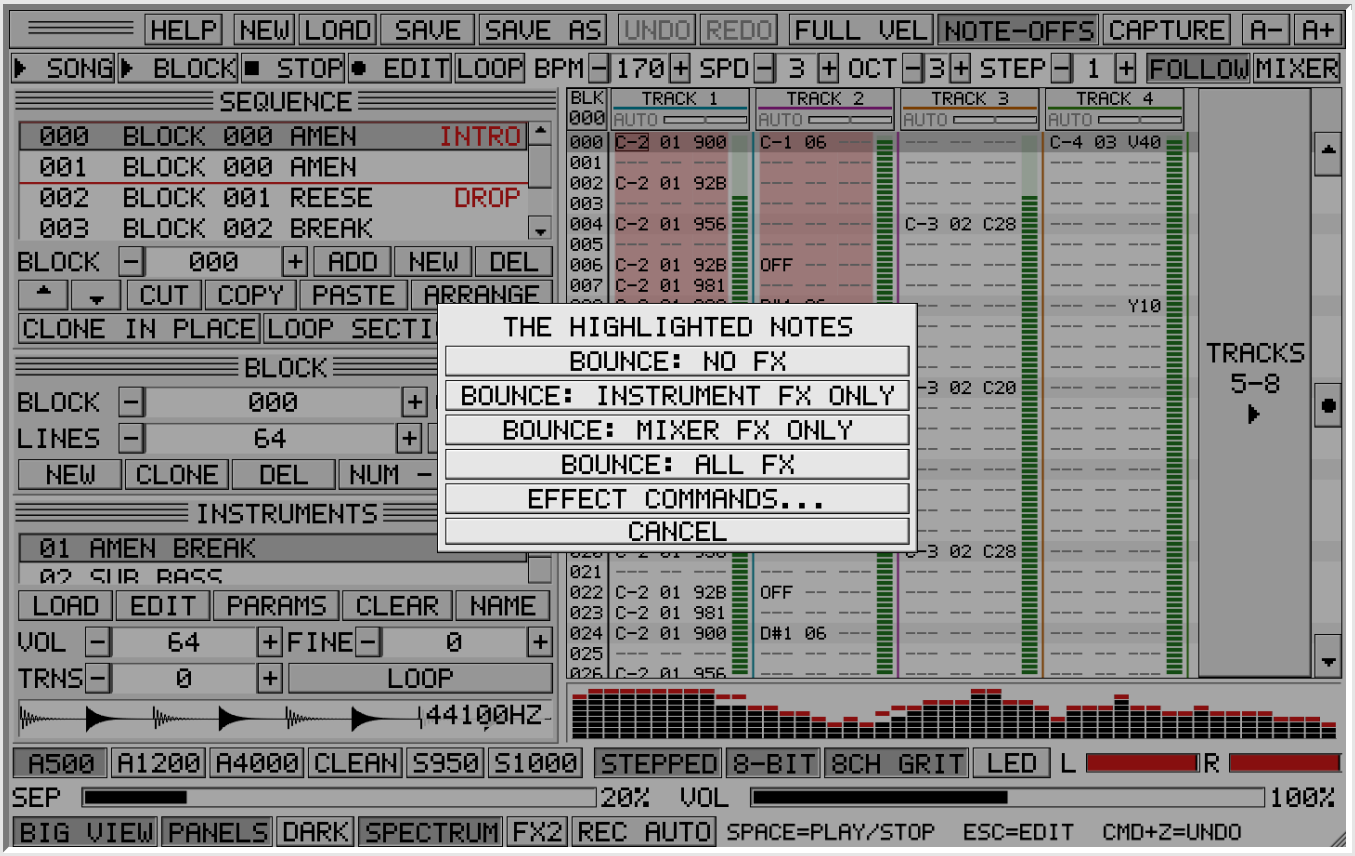
116 chord types - major, minor, dominant, suspended, diminished, augmented and more - from the tonal music library's list, which comes from Wikibooks' complete list of chord patterns. Each one is there as written (ROOT), in every inversion (INV 1, INV 2…), as DROP 2 and DROP 3 voicings for four-note chords and OPEN for triads: 638 voicings in all.

THE SONG'S CHORDS AND THE GLOBAL SET

- `EVERY SONG HAS ITS OWN TABLE`, saved in the song. The `GLOBAL SET` is kept on your Mac: every new song, and every MOD or older song without a table, starts with a copy of it.
- Changing a song's chords doesn't change the global set, and changing the global set doesn't change any song. A saved song always plays its own chords, whatever the global set has become.
- When you load a song, the status line says how its chords compare with the global set, for example `SLOTS 05 1F DIFFER FROM THE GLOBAL SET`. It plays its own either way.
- Older songs are converted when they load: v0.12–v0.14 chords (`0xy` on a CHORD instrument) become `#xy`, and v0.11 chords (`0xy` as two intervals) each get a slot with the same notes. The song sounds the same.

BOUNCING

Turn any part of the pattern into a new sample. Highlight some notes (they can cover several tracks), then right-click inside the highlight, or press CMD B.



RIGHT-CLICK INSIDE A HIGHLIGHT

CHOICE	WHAT'S IN THE SAMPLE
NO FX	The instruments' sound before any effects. Use it when the sample will play through the same effects.
INSTRUMENT FX ONLY	Through the instruments' own inserts, not the mixer.
MIXER FX ONLY	After the tracks' strips (EQ, inserts, fader) plus what they send to SEND A / B, so reverb and delay tails are in it; not the instruments' own effects.
ALL FX	Both. Never the master or the Paula / Akai output stage. The new instrument is called BOUNCE, BOUNCE+IFX, BOUNCE+MFX or BOUNCE+FX.

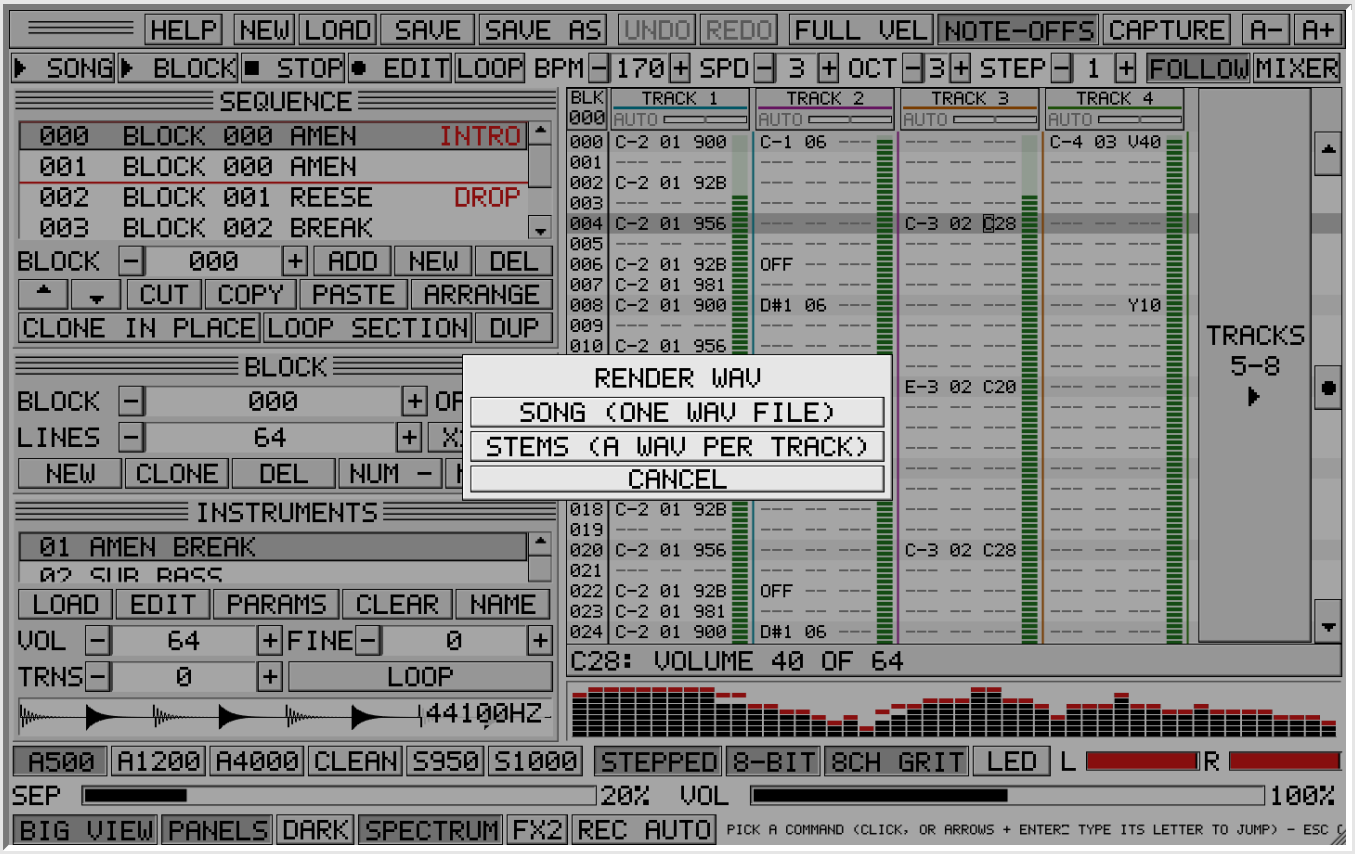
- Only the highlighted lines of the highlighted tracks play, from the first line, at the tempo and speed set there. Plug-in

instruments are included.

- It records on after the last line until the sound has died away (up to 10 seconds), so tails aren't cut.
- The sample goes into the first empty instrument slot, named like `BOUNCE T1-2 000-015`, at 44.1 kHz. `C-3` plays it exactly as recorded.
- Nothing else changes: your pattern and the instruments that played it stay as they are. Undo removes the new instrument.

RENDERING

`RENDER WAV` (File menu) asks what to render:



SONG OR STEMS

- **SONG:** the song once through as a 24-bit stereo WAV, mixer and all.
- **STEMS:** a folder next to the name you pick, `<name> STEMS`, with one 24-bit WAV per used track (after its strip), SEND A, SEND B and the master: `01 TRACK 1.wav` … `11 MASTER.wav`. They start together and are the same length, so they drop straight into Ableton.
- After the song ends it keeps going until everything is quiet for a second (up to 20 seconds), so tails aren't cut. Plug-in delay is taken off the start.
- MIDI-out instruments can't be rendered; the status line says which were left out.

These are in the **File** menu. `EXPORT SAMPLES` writes every instrument's sample to a folder as `01 NAME.wav`, `02 NAME.wav` …

SAVE MOD writes a ProTracker MOD (M.K. when only tracks 1-4 are used, else 8CHN) and says what a MOD can't hold.

SAVING & FILES

- SAVE (CMD S) writes straight to the song's file; the first time it asks for a name. SAVE AS (CMD SHIFT S) always asks, for versions. The song takes the file's name. SAVE* with a star means unsaved changes.
- A .prv song holds everything: every sample (stereo too), the mixer, plug-in settings, automation, MIDI-out settings, its chord table. You can move or delete the original WAVs.
- LOAD also opens ProTracker-style .mod files (4, 6 and 8 channels) and MED / OctaMED MMD0 / MMD1 modules.
- Songs saved by older versions always load.

KEYS

KEY	ACTION
Z S X D C…	Notes, lower octave (, L . ; / carry on up)
Q 2 W 3 E…	Notes, upper octave
1	Note off
SPACE	Play / stop the block · with SHIFT, the song. On every page: Space stops the song
ESC or \	Edit mode on / off
SHIFT 3	In a command column: #, the chord command
ENTER	Play block · SHIFT song · CMD block from the cursor
Arrows	Move · SHIFT ↑ ↓ 4 lines · SHIFT ← → track
TAB	Next track · with SHIFT, previous
[ ]	Octave down / up, on every page (also with a plug-in's window in front)
- =	Previous / next instrument
DELETE	Clear the highlight or the field · SHIFT removes the line · ALT inserts one
CMD C X V	Copy / cut / paste the highlight, or the track (SHIFT: the block)
CMD A	Highlight the whole block
CMD B	Bounce the highlight to a new instrument
CMD ↑ ↓	Transpose the track a semitone (SHIFT: an octave)
CMD Z	Undo · with SHIFT, redo
CMD [ ]	Loop start / end at the cursor
CMD S	Save · with SHIFT, save as
CMD 2 / 1	Mixer / pattern
CMD 3	CHORDS: the chord table
CMD 4	SAMPLE: record audio into the instrument

KEY	ACTION
CMD F	FOLLOW on / off
CMD R	CAPTURE: write what you just played into the pattern
CMD O	Load a song
ALT ↑ ↓	Move the selected song positions